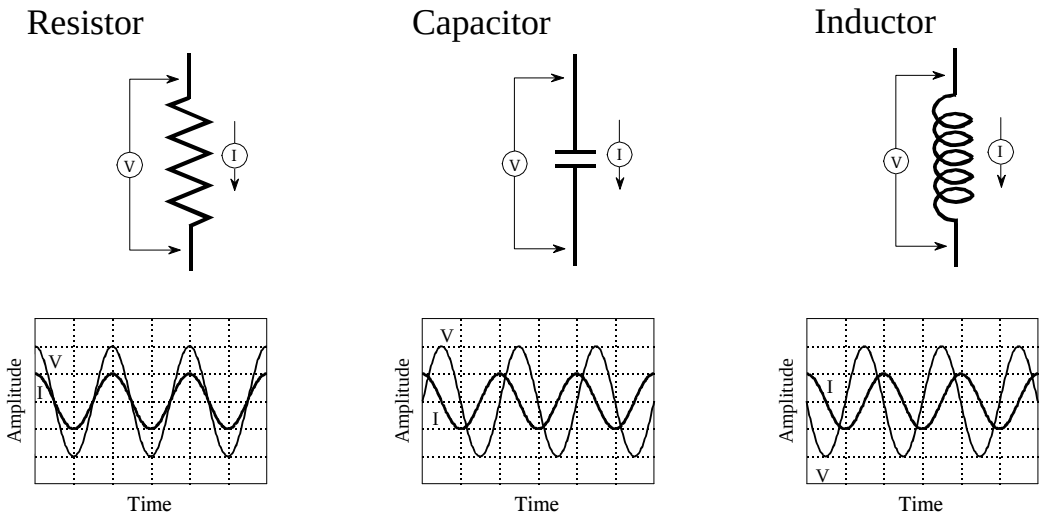

FIGURE 30-5
Definition of impedance. When sinusoidal voltages and currents are represented by complex numbers, the ratio between the two is called the *impedance*, and is denoted by the complex variable, Z. Resistors, capacitors and inductors have impedances of R, $-j/\omega C$, and $j\omega L$, respectively.

found. In polar form, the magnitude of the impedance is the ratio between the amplitudes of V and I. Likewise, the phase of the impedance is the phase difference between V and I.

This relation can be thought of as *Ohm's law for sinusoids*. Ohms's law ($v = iR$) describes how the *resistance* relates the instantaneous current and voltage in a resistor. When the signals are sinusoids represented by complex numbers, the relation becomes: $V = IZ$. That is, the *impedance* relates the current and voltage. Resistance is an ordinary number, since it deals with two ordinary numbers. Impedance is a complex number, since it relates two complex numbers. Impedance contains more information than resistance, because it dictates both the amplitudes *and* the phase angles.

From the differential equations that govern their operation, it can be shown that the impedance of the resistor, capacitor, and inductor are: R, $-j/\omega C$, and $j\omega L$, respectively. As an example, imagine that the current in each of these components is a unity amplitude cosine wave, as shown in Fig. 30-5. Using substitution, this is represented by the complex number: $1+0j$. The voltage across the resistor will be: $V = IZ = (1+0j)R = R+0j$. In other words, a cosine wave of amplitude R. The voltage across the capacitor is found to be: $V = IZ = (1+0j)(-j/\omega C)$. This reduces to: $0-j/\omega C$, a sine wave of amplitude, $1/\omega C$. Likewise, the voltage across the inductor can be calculated: $V = IZ = (1+0j)(j\omega L)$. This reduces to: $0+j\omega L$, a negative sine wave of amplitude, ωL.

The beauty of this method is that *RLC* circuits can be analyzed without having to resort to differential equations. The *impedance* of the resistors, capacitors,